

Jaiaditya Asudani

Github: github.com/Jai2379
Portfolio: [Jai2379.github.io](https://jai2379.github.io)

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EDUCATION

- University of Waterloo** Waterloo, ON
 - Candidate for BAsC in Nanotechnology Engineering* *September 2025 – Present*
 - Expected Graduation: May 2030*
 - Relevant Coursework: Materials Science & Nanomaterials, Electricity, Magnetism & Circuits, Computational Methods*

SKILLS SUMMARY

- Computational Tools:** Python (NumPy, SciPy, Streamlit), MATLAB, C/C++ (including Arduino), FEA
- Design & Analysis:** SolidWorks, AutoCAD, CATIA, Fusion360, OnShape, FEA, GD&T
- Hardware:** Micro-controllers, Embedded Sensors, Circuit Design, Machining, Prototyping
- Soft Skills:** Technical Problem-Solving, Time Management, Team Collaboration, Attention-to-Detail

EXPERIENCE

- Waterloop (Design Team) — University of Waterloo** Waterloo, Ontario
 - Mechanical Engineering Member* *October 2025 - Present*
 - Full-Scale Hyperpod Assembly:** Integrated 20+ mechanical components into a full-scale hyperpod assembly using SolidWorks and Onshape, resolving fit, alignment, and subsystem interface constraints.
 - Mechanical Design & CAD:** Designed and iterated 3 mechanical components across multiple revision cycles, applying DFM principles for manual machining (mill, lathe) and basic CNC fabrication.
 - Materials Research (Carbon Fiber):** Conducted carbon fiber composite materials research, evaluating strength-to-weight tradeoffs to support lightweight structural design decisions. 7% reduction in weight.
- Electrium Mobility (Design Team) — University of Waterloo** Waterloo, Ontario
 - Mechanical Engineering Member* *October 2025 - Present*
 - System-Scale Development:** Modeled and assembled structural and mounting components for a full-scale electric bike in SolidWorks, contributing to system-level mechanical integration.
 - Interface Evaluation:** Designed and integrated 3+ human-machine interface components (throttle housing, handlebar elements, display mounts) with emphasis on ergonomics and durability.
 - Practicality:** Applied design-for-manufacturability (DFM) principles, accounting for tolerance stack-up, fastener access, and assembly sequencing to improve prototype reliability.
- Mathematics Tutor** Brampton, Ontario
 - Private/Independent* *February 2025 - May 2025*
 - Mentorship:** Delivered weekly 1:1 tutoring sessions over 4+ months, resulting in a 13% average improvement in students' mathematics grade.
 - Personalization:** Developed custom lesson plans to strengthen problem-solving and logical reasoning in advanced mathematics.
 - Management:** Independently managed lesson planning, scheduling, and progress tracking across a multi-month timeframe.

PROJECTS

- TENG Device (Tribo-Electric Nano Generator):** Designed and fabricated a triboelectric nanogenerator (TENG) during a 2-day rapid prototyping challenge, achieving 22 V electrical output as a proof-of-concept. — *SolidWorks, Engineering-Design*
- 6-DOF Robotic Arm (Prototype):** Engineered a 6-degree-of-freedom robotic arm in SolidWorks, developing a fully articulated mechanical assembly with defined motion constraints. Designed for modularity and scalability, enabling future Arduino-based actuation and control integration. — *SolidWorks, 3D Printing*
- Adaptive Presence-Based Lighting Control System (WIP):** Developing a smart lighting system using an Arduino Uno and PIR motion sensor to dynamically regulate lighting based on user presence. Implementing embedded C++ logic and digital I/O control to improve energy efficiency during inactivity. — *Arduino, Motion-Detection, C++, Circuits*
- Interactive Data Analysis Tool:** Developed an interactive histogram and implemented dynamic binning, distribution fitting, and user-controlled parameters to support exploratory data analysis and statistical insight. — *Python, NumPy/SciPy, Streamlit*

VOLUNTEER EXPERIENCE

- Royal Canadian Air Cadet Squadron** Brampton, Ontario
 - Taught and led a batch of 30+ students on aviation and leadership.* *September 2020 - June 2025*
- IGNITE Stem** Brampton, Ontario
 - Made STEM more accessible for students, held hackathons and other events.* *September 2025 - June 2025*